

Potential Modern AutoCrit Project Proposals

Contents

I	Main Idea	2
1	1. AutoCrit MetricSpace: Quantifying Eligibility and Population Similarity	2

Part I

Main Idea

1 1. AutoCrit MetricSpace: Quantifying Eligibility and Population Similarity

Modern Autocrit already extracts normalized eligibility criteria, but the current output mainly supports qualitative comparisons: it can indicate that two criteria are different without quantifying *how different* they are. This project would develop a statistically meaningful distance measure between individual eligibility claims and between the trial populations those claims define. The resulting distance could be used to retrieve comparable trials, determine how much historical information to borrow, and quantify how proposed eligibility changes propagate into sample size, power, recruitment, and generalizability.

Criterion-level distance

A general-purpose sentence embedding can represent the semantic meaning of two claims and compare their vectors using cosine or Euclidean distance. This is useful for recognizing that “age at least 60” and “participants must be 60 years or older” express the same requirement. However, embeddings alone may assign excessive similarity to clinically different thresholds such as “age ≥ 60 ” and “age ≥ 18 ” because nearly all of their words are identical.

The proposed system would therefore use a hybrid representation containing:

- a sentence embedding for the criterion’s semantic meaning;
- structured fields produced by Modern AutoCrit, including entity, attribute, comparator, threshold, unit, temporality, negation, and logical relationships;
- domain-aware numerical distances that account for the clinical scale and direction of a threshold; and
- an LLM or trained classifier for relationships that cannot be reliably resolved from vectors alone.

A criterion distance could take the form

$$d(c_i, c_j) = w_s d_{\text{semantic}}(c_i, c_j) + w_a d_{\text{attribute}}(c_i, c_j) + w_n d_{\text{numeric}}(c_i, c_j) + w_l d_{\text{logic}}(c_i, c_j),$$

where the weights are learned or calibrated against human judgments and downstream performance. For numeric criteria, the distance should use a clinically meaningful scale rather than raw token similarity. For example, thresholds could be standardized by an observed population distribution, reference range, or robust scale so that the model recognizes the practical difference between age cutoffs of 18 and 60.

LLM-assisted distance quantification

For criterion pairs that remain ambiguous after structured parsing, an LLM could act as a constrained pairwise evaluator. Given two source claims and their extracted fields, it would classify their relationship—for example, equivalent, narrower, broader, partially overlapping, contradictory, or unrelated—and assign an ordinal distance with a short evidence-based explanation. The LLM score would be calibrated against reviewed criterion pairs and combined with, rather than

substituted for, the embedding and structured numerical distances. Repeated evaluations or multiple models could provide an uncertainty measure, with high-disagreement pairs routed to human review.

- Require the LLM to compare the clinical effect of changed thresholds, units, negation, time windows, and logical operators rather than relying on surface wording.
- Evaluate agreement with expert pairwise rankings, calibration of the proposed distance, and consistency across repeated prompts and model choices.
- Use the LLM-derived relationship as an interpretable feature in the final hybrid metric and preserve its explanation for auditing trial matches.

Population-level distance

After defining criterion-level costs, compare two trials as collections of eligibility constraints. One approach is optimal transport: each trial is represented as a weighted set of criteria, and the transport plan matches related criteria across protocols while paying a higher cost for unmatched or substantially different requirements.

$$D(P, Q) = \min_{\gamma \in \Pi(P, Q)} \sum_{i,j} \gamma_{ij} d(c_i, c_j).$$

This produces both an overall distance and an interpretable alignment showing which criteria make the populations different. If patient-level or synthetic population data are available, the project could also compare the distributions of patients who pass each protocol using Wasserstein distance, energy distance, or maximum mean discrepancy. The text-level and patient-level measures can then be compared to determine whether semantic protocol differences correspond to meaningful population shifts.

How far should the system pull comparable trials?

The retrieval radius should be selected empirically rather than chosen only from visual similarity. The project would retrieve historical trials at increasing distance thresholds and evaluate the bias–variance tradeoff:

- a narrow radius returns highly similar trials but may provide too little historical information;
- a wide radius increases the available sample but may introduce nonexchangeable populations and biased assumptions; and
- the preferred radius minimizes a prespecified downstream loss, such as prediction error, prior-data conflict, treatment-effect bias, or poor confidence-interval coverage.

Cross-validation or simulation can select the distance threshold, with uncertainty reported through bootstrap intervals. In a Bayesian design, distance could control the degree of historical borrowing, but the mapping from distance to prior weight must be prespecified and stress-tested under population conflict.

Propagation into trial design

The distance measure becomes useful when connected to concrete design decisions. The system could:

- retrieve historical trials for control-rate, variance, dropout, or recruitment assumptions;
- discount historical effective sample size as population distance increases;
- estimate how relaxing one criterion changes the target population distribution;
- update expected event rates, outcome variance, required sample size, and power under the shifted population; and
- show which individual criteria account for most of the distance between two protocols.

Recommended modeling architecture

A two-stage retrieval and reranking architecture is more practical than expecting one embedding to solve every comparison:

1. **Bi-encoder retrieval.** Encode every criterion independently and use cosine similarity to retrieve a small set of potentially related criteria. This stage is fast enough to search thousands of protocols.
2. **Cross-encoder comparison.** Pass each retrieved pair jointly through a second model that can attend directly to differences in thresholds, comparators, negation, units, and time windows.
3. **Structured distance layer.** Combine the learned scores with deterministic distances computed from Modern AutoCrit’s extracted fields.
4. **Protocol aggregation.** Use optimal transport to align the weighted criterion sets and calculate an overall protocol or population distance.

The pairwise model should predict more than a single similarity number. A useful multi-task output is:

- semantic relationship: equivalent, narrower, broader, partial overlap, contradictory, or unrelated;
- criterion type and matched attribute;
- continuous distance on a calibrated scale;
- direction of population inclusion, where applicable; and
- uncertainty or an abstention decision.

For example, “age ≥ 60 ” is narrower than “age ≥ 18 ,” while “age < 60 ” partially overlaps with neither in the same way. A directional label is therefore more informative than a symmetric cosine score. The final distance can remain symmetric for optimal transport while retaining the directional relation as metadata.

Hugging Face starting models

The project should benchmark several starting checkpoints rather than assume that biomedical pretraining will automatically solve criterion comparison:

- `FremyCompany/BioLORD-2023` is the strongest first candidate for the bi-encoder. It is already packaged for Sentence Transformers and was trained to represent biomedical concepts and clinical sentences. Its specialized terminology licenses should be reviewed before any commercial use.
- `ncbi/MedCPT-Query-Encoder` is a public-domain biomedical retrieval baseline trained on large-scale PubMed search behavior. It supports short-text and query-to-query representations, making it suitable for criterion retrieval.
- `cambridge/t1/SapBERT-from-PubMedBERT-fulltext` is useful for the entity and terminology component of the distance. Because it was developed for biomedical concept representation and entity linking, it should not be treated as a complete sentence-level relation model by itself.
- `microsoft/BiomedNLP-BiomedBERT-base-uncased-abstract-fulltext` is a good base checkpoint for a custom cross-encoder that classifies the relationship between two criteria. It provides biomedical language pretraining without imposing an existing sentence-similarity objective.
- `ncbi/MedCPT-Cross-Encoder` is an additional biomedical reranking baseline. It was not trained specifically for symmetric eligibility comparison, so the classification head should be replaced and fine-tuned on criterion pairs.
- `all-mpnet-base-v2` and `bge-base-en-v1.5` provide strong general-domain baselines. If a biomedical model does not outperform them, that is an important experimental result.

A sensible initial comparison is BioLORD versus MedCPT versus MPNet for retrieval, followed by BiomedBERT as the pairwise cross-encoder. This keeps all primary models near the BERT/base scale and makes fine-tuning feasible on a single modern GPU. Parameter-efficient fine-tuning can be used if an instruction-tuned LLM is later added as a pairwise judge, but the encoder models above are more suitable for producing a scalable metric space.

Training-data construction

Modern AutoCrit can create the initial corpus by extracting and normalizing criteria from ClinicalTrials.gov protocols. Training examples would consist of criterion pairs, structured fields, and reviewed relationship labels. The dataset should deliberately include difficult contrasts rather than mostly easy unrelated pairs:

- **Equivalent positives:** paraphrases, abbreviations, terminology synonyms, and unit-converted expressions that define the same population.
- **Numeric hard negatives:** identical wording with different age, laboratory, dosage, score, or duration thresholds.
- **Logical hard negatives:** comparator reversal, inclusion versus exclusion, negation, AND/OR changes, and exceptions.
- **Temporal hard negatives:** the same event with different lookback, washout, or stability periods.
- **Nested clinical differences:** the same disease with different stage, biomarker, treatment line, disease activity, or organ-function requirements.
- **Unrelated negatives:** criteria from different attributes or clinical domains.

An LLM can provide weak labels and explanations at scale, but a manually reviewed subset should define the evaluation standard. Synthetic perturbations are useful because they create controlled threshold and logic changes, but the test set should contain naturally occurring criteria from protocols. Data should be split by trial—and preferably also tested on held-out disease areas—to prevent near-duplicate criteria from leaking across training and evaluation.

Fine-tuning objectives

The bi-encoder can be trained with contrastive or triplet loss. An anchor criterion should be closer to an equivalent or near-equivalent criterion than to a clinically different hard negative:

$$L_{\text{triplet}} = \max\{0, d(c_a, c_p) - d(c_a, c_n) + m\}.$$

The cross-encoder can use a multi-task loss combining relationship classification, continuous-distance regression, and directional ordering:

$$L = \lambda_1 L_{\text{relation}} + \lambda_2 L_{\text{distance}} + \lambda_3 L_{\text{direction}} + \lambda_4 L_{\text{contrastive}}.$$

Structured numeric features should either enter a small calibration network alongside the text representation or be combined with model scores after training. Keeping the structured component explicit makes failures such as the age-threshold example easier to diagnose and audit.

Evaluation plan

Evaluation should occur at three levels:

- **Pair level:** macro F1 for relationship classes, Spearman correlation with expert distance rankings, triplet-ordering accuracy, calibration error, and performance on numeric, negation, unit, and temporal challenge sets.
- **Retrieval level:** Recall@ k , mean reciprocal rank, and normalized discounted cumulative gain for recovering genuinely comparable criteria and trials.
- **Design level:** error in predicted historical control rates or variances; prior-data conflict; treatment-effect bias; interval coverage; and sample-size or recruitment error as the retrieval radius changes.

Important ablations include embedding only, structured fields only, LLM judgment only, bi-encoder plus structured distance, and the complete bi-encoder–cross-encoder–structured model. This will show whether the added complexity produces clinically and statistically meaningful improvement.

Course-project sketch

- Extract a corpus from one or two therapeutic areas and create approximately 2,000–5,000 weakly labeled pairs, including controlled numeric and logical perturbations; manually review a smaller validation and test set.
- Benchmark BioLORD, MedCPT, and MPNet as frozen bi-encoders, then fine-tune the strongest model with hard-negative contrastive learning.

- Fine-tune a BiomedBERT cross-encoder to predict pairwise relationship, direction, and distance, and combine it with Modern AutoCrit’s structured numeric and logical features.
- Aggregate criterion costs into a protocol-level optimal-transport distance and evaluate retrieval of comparable trials.
- Demonstrate one downstream use case: vary the retrieval radius or distance-weighted borrowing and show its effect on a simulated sample-size, power, recruitment, or historical-control calculation.
- Build an interface that aligns two protocols, highlights the criteria responsible for their distance, exposes model uncertainty, and routes ambiguous comparisons for review.